HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and developed **Java Spring Boot** microservices for backend orchestration, implementing RESTful APIs and handling complex business logic.
- Integrated **large language models** into backend services to generate structured, machine-readable plans from natural language inputs, managing non-deterministic outputs.
- Developed Python scripts to automate ETL processes for data pipelines, transforming and loading data from various sources into a common data model for downstream AI agent consumption.
- Owned end-to-end feature development for the Agentic Orchestration system, from initial design and **API specification** through deployment in an agile team environment.
- Collaborated with multiple stakeholders, including product managers and other engineering teams, to gather requirements and align on technical solutions for AI-powered experiences.
- Participated actively in code reviews, providing constructive feedback and ensuring adherence to **OOP principles** and design patterns for a 7-person engineering team.
- Containerized backend services using **Docker** and deployed them to a cloud environment, ensuring efficient resource utilization and enabling seamless scaling.

SKILLS

Languages – Java, Python, SQL, R.

Frameworks & Libraries – Spring Boot, Flask, FastAPI, pandas, NumPy, scikit-learn, LangChain.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, Git.

Databases – PostgreSQL, MySQL, Oracle.

AI/ML Concepts – LLM Integration, Generative AI, Prompt Engineering, Structured Output Generation, Non-deterministic System Handling, Agentic Orchestration, Model Evaluation.

Methodologies – Agile, Scrum, RESTful API Design, Microservices, OOP, Design Patterns.

PROJECTS

AI Agentic Orchestration Backend Service

- Built a **Java Spring Boot** backend service to manage the orchestration of AI agents, interpreting customer intent and deciding the next best action.
- Integrated **large language models** to process natural language requests, generating structured plans and validating them against defined schemas for runtime execution.
- Developed RESTful APIs for seamless integration with internal systems, normalizing data from diverse sources into a unified model for agent decisioning.

Natural Language to Action Plan Generator

- Engineered a **Python**-based service that translates user-defined natural language goals into executable plans for an AI orchestration system, handling edge cases and retries.
- Implemented schema enforcement for generated plans to ensure completeness and consistency, reducing runtime errors by approximately **30%**.
- Integrated with a **vector database** to retrieve relevant context and building blocks, enriching the plan generation process with contextual information.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)